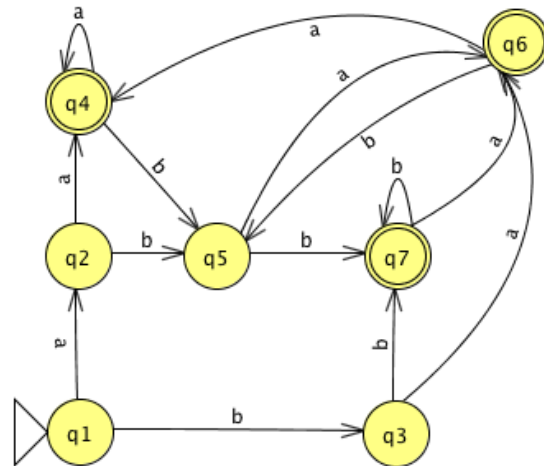


1. Minimize the following finite automaton: $(\{q1, q2, \dots q7\}, \{a, b\}, \delta, q1, \{q4, q6, q7\})$, where δ is as follows:

	a	b
q1	q2	q3
q2	q4	q5
q3	q6	q7
q4	q4	q5
q5	q6	q7
q6	q4	q5
q7	q6	q7

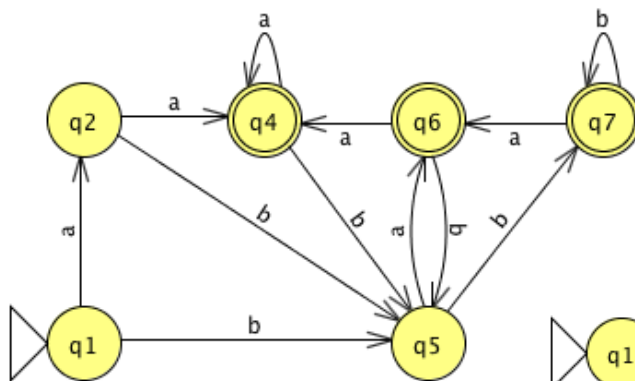


Minimizing DFA by Combining States:

q2	X					
q3	X	X				
q4	X	X	X			
q5	X	X	O	X		
q6	X	X	X	O	X	
q7	x	X	X	X	X	X
	q1	q2	q3	q4	q5	q6

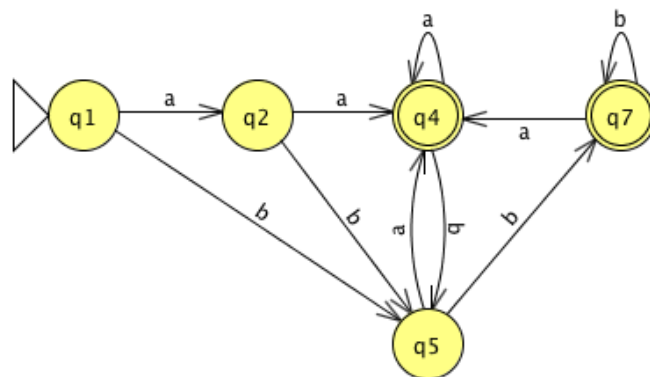
We found that $q5=q3$ and $q6=q4$, removing $q3$ and $q6$ from the original DFA and matching transitions to each one of the removed equivalence state.

Removing $q3$, transition on "b" from $q2$ to $q3$ is now from $q2$ to $q5$.



Next, removing $q6$, transition on "a" from $q7$ to $q6$ is now from $q7$ to $q4$.

Resulting Minimized DFA:



2. Problem 1.53: Let $\Sigma = \{0,1,+,=\}$ and let $ADD = \{x=y+z \mid x \text{ and } y \text{ and } z \text{ are binary integers and } x = y+z\}$.

Prove that ADD is not regular using the pumping lemma.

Assume, for a contradiction, that ADD is regular. Then, ADD has a length $m > p$ where p is the pumping length. Now, let's choose s to be a string that is in the language and that will meet the pumping lemma requirements where:

- a) $|y| > 0$
- b) $|xy| \leq p$
- c) $xy^iz \in ADD$ for all $i \geq 0$

Let $s = 1^p$, since $x = y + z$ we can say that $s = 1^p = 1^p + 0$.

Now from a and b we have that $y = 1^k$ for some $k > 0$ and $k \leq p$. According to the pumping lemma we can choose any i , and the string s will still be in the language. Let's choose $i = 2$, then $xy^2z = 1^{p+k}$. This says that $1^{p+k} = 1^p + 0$ which $\notin ADD$ since it does not satisfy binary addition, this is a contradiction.

3. Give a context-free grammar for the following language:

$\{0^n 1^m \mid n \neq m\}$

$S \rightarrow 0A \mid B1$

$A \rightarrow 0A1 \mid 0A \mid \varepsilon$

$B \rightarrow 0B1 \mid B1 \mid \varepsilon$

4. Give a PDA for the same language as in the last problem.

